InterActions

across physics and education

September/October 2007

Focal Point

Physicists at Work

Inside:

Career Paths, Profiled

Physics Majors, Uncovered

Writing About Science

and introducing "Homer Dodge's Notebook"

InterActions
across physics and education

About INTERACTIONS
Interactions is a general-interest magazine about physics education. Our mission is to inform and stimulate diverse conversations on teaching and learning by publishing thought-provoking news, analysis, and commentary on the people, programs, and policies that interact to influence scientific practices and knowledge—and, ultimately, human destiny.

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The editors welcome your response. Send comments, questions or suggestions to interactions@aapt.org or mail letters to Interactions Forum, One Physics Ellipse, 5th Floor, College Park, MD 20740. Please include your full name, mailing address, and daytime contact information. Space is limited and all published comments are subject to editing.

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Graduates toss their mortar boards high into the air during graduation ceremonies hoping they are prepared for the real world.
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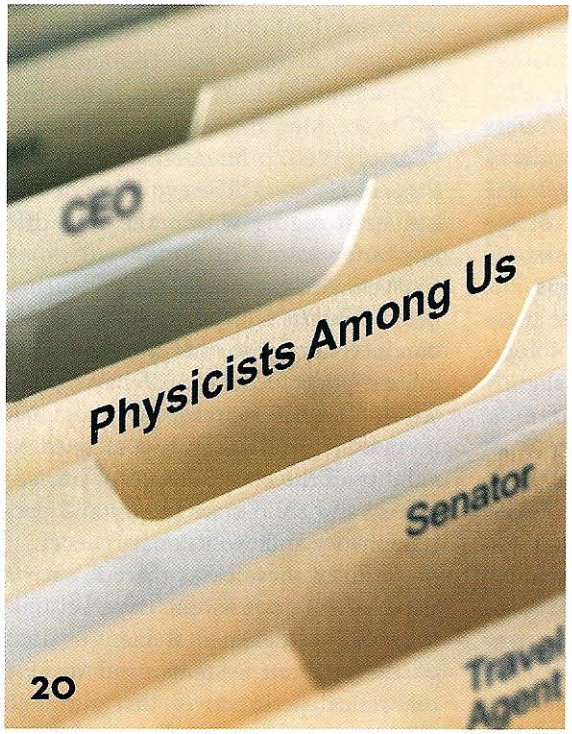

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Physics and Faith

In an era when science is argued to be the antithesis of religion, however shall the twain meet? Answer: Father Cyril Opeil, physicist and Jesuit priest. An assistant professor of physics at Boston College, Father Cyril reconciles faith and science on a daily basis.

Father Opeil was a young engineer working on aircrafts for Westinghouse in Baltimore in the 1980s before being called to the priesthood. He chose the Jesuits because of their well-known scholarly tradition and spiritual devotion. "It was a wonderful combination that the Jesuits offered," Opeil says. "I could have a profession, and be a priest and live a ministerial life."

Such a creed offered the scholar in Opeil the perfect opportunity to devote himself monastically to his work. A few years after his ordination, Opeil taught chemistry and physics at Gonzaga High School, a distinguished Jesuit school in Washington, D.C. His provincial, Rev. Edward Glynn, recognized his scientific aptitude and suggested he pursue doctoral studies.

Opeil earned a Ph.D. in physics at Boston College. He then accepted a two-year postdoc, becoming the first Jesuit priest to work at Los Alamos National Laboratories.

While at Los Alamos, Opeil attached himself to the tiny parish of Chimayo, which is tucked away in the Sangre de Cristo Mountains. Although encouraged by his superiors to exempt himself due to his research schedule, he took on a full range of ministerial duties—saying Masses, hearing confessions, conducting weddings and baptisms.

The heart of the parish is the Sanctuaria de Chimayo, a small chapel also known as the "Lourdes of America," where pilgrims come to carry away small amounts of the red earth to which they ascribed healing properties.

Founded in 1534 by Ignatius of Loyola, the Society of Jesus boasts a long history of scholarly and intellectual achievement. The Jesuit scientists that Opeil counts among his precursors include the polymath Athanius Kircher, the German astronomer Christopher Clavius, and the French paleontologist Teilhard de Chardin. Opeil point outs that 23 lunar craters are named after Jesuits. "We were the people mapping the moon."

"It was the idea of our founder, Ignatius, to educate and nurture the whole person, to care for the soul, not just to make sure people are meeting certain requirements," Opeil says.

Much of his research involves studying electrons in uranium. Opeil unifies the supposed dichotomy of his two callings through the Jesuit motto, "To find God in all things," as he unlocks the mysteries of the smallest components of the natural order. Says Father Opeil, "Science is a way of looking at how the Creator manifests himself in the world." Δ